blender — LEARN WITH A POSTER

EVERYTHING YOU NEED IN A SINGLE MAP

VER. 29/08/2024 FOR BLENDER 4.2 THIS MAP REFLECTS A PERSONAL MIND MAP AND EXPERIENCE IN USING BLENDER 3D.
WILL IMPROVE AND UPDATE FOLLOWING THE DEVELOPMENT AND SKILLS. MADE BY GIUDANSKY.COM, FOLLOW ME FOR UPDATES.

MODEL · SELECT · MANIPULATE · UI · ORGANIZE · PAINT TEXTURE POST PROD · PLUGINS · MISC · new 4

SIZE and **thickness** define **IMPORTANCE**!

Cursor, origin, orientation

FIRST THING YOU NEED TO LEARN: ADD A MONKEY HEAD ON TOP OF THAT WONDERFUL CUBE!

Shift+A **Add** object — or X / CANC **delete** object

Ctrl+X dissolve

CURSOR — VERSATILE REFERENCE POINT IN 3D SPACE.

Shift+Cursor place

Shift+S **relocate** selection/cursor

Shift+C reset 3d cursor to origin

transform **orientation** — TRANSFORM RELATIVE TO GLOBAL, LOCAL, NORMAL, GIMBAL, VIEW, CURSOR. TIP: SELECT AND ... FOR CUSTOM ORIENTATION

Select

select

W select mode

object

Shift+add to selection

border select

Ctrl+lasso select

C brush select — brush size

A **Select all** / Alt+A deselect

Mesh select mode face, edge, vertex — 1 / 2 / 3

Ctrl+nvert select

Ctrl+ expand / reduce selection

Shift+G select grouped / select similar

Alt+ select menù — LIST AND SELECT OBJECTS BEHIND THE CURSOR

Alt+Q switch select object on hover

Ctrl+ select linked

Alt+ loop select

Ctrl+Alt+ ring select

Alt+ select face

Ctrl+ shortest path select

Ctrl+Alt+ interval select

Transform

Grab (move) — ×2 slide along vertex or edges & C slide over

&x/y/z **axis** constrain — ×2 switch orientation — &Shift+x/y/z lock axis

&HOLD Ctrl snap to units

&HOLD Shift precision mode

alt+ reset transform

Scale — SCALES ALONG SELECTED NORMALS — alt+ Shrink - fatten

Affect only: origins — TRANSFORM ORIGINS, LOCATIONS, PARENTS...

Ctrl+Mirror — ORIGIN — OBJECT'S ORIGIN ACTS AS THE PIVOT POINT FOR TRANSFORMS

f9 adjust last — REFINE PARAMETERS OF THE LAST OPERATION

Shift+Repeat last — VERY USEFUL FOR REPETITIVE ACTIONS

Rotate ×2 free rotation

Shift+Duplicate — DIFFERENT OBJECT DIFFERENT DATA

Alt+Duplicate linked — DIFFERENT OBJECT SAME DATA

extend vertices — ADD VERTICES ALONG THE EDGE IN THE DIRECTION OF THE MOUSE CURSOR

Ctrl+Apply transform — FIX STRANGE BEHAVIOURS IN EDIT MODE AND MODIFIERS. BE CAREFUL, YOU'LL NOT BE ABLE TO RESET TRANSFORMS BY NUMERIC VALUES

Ctrl+Copy attributes — COPY TRANSFORM, UV AND MODIFIERS, FROM ACTIVE TO SELECTED

ctrl+A Skin resize

3d viewport

Z Viewport shading — wire · solid · lookdev · render

/ **Isolation** mode on selected — FOCUS ON A SINGLE MESH EDITING

Shift+ Hide unselected

Hide selected

alt+ unHide

Zoom

Shift+B crop zoom

Shift+ **Pan**

alt+Z X-ray view

alt+B Isolation crop

Views

7 top · 8 · 9 opposite · Ctrl+ opposite view

4 · ortho 5 · 6

1 front · 2 · 3 right · Shift+ align to active

camera 0 user

view on selected — ROTATION AND ZOOM CENTER ON OBJECT AND SCULPT AREA

Search

f3 **Search** — CAN'T FIND SOMETHING? SEARCH IT! HE LIKE IN CONVERSATION WITH QUICK FAVOURITES TO IMPROVE UI WORKFLOW

spacebar **Search** in menus

f1 **Blender wiki** — CLICK WHILE HOVERING, OPEN HELP PAGE WITH INFOS ON ACTUAL ELEMENT

Ctrl+ Workspaces

Quick favourites

add/change shortcut

add to quick favourites

UI

T **toolbar** panel

N **sidebar** panel — item, tool, view, plugins

divide view — RMB: SPLIT VERTICAL, SPLIT HORIZONTAL

maximize window — Ctrl+spacebar

modes Pie menu

Ctrl+ **tab** — EDIT / OBJECT MODE, INTERACTION MODE

object · edit · weight paint · vertex paint · texture paint · sculpt · draw

Ctrl+alt+Q — 4 sides view: TOP, USER, FRONT, RIGHT — OLDSCHOOL ORTHOGRAPHIC PROJECTIONS. THE GOOD OLD 3DS TIMES...

alt+ change num value to multi objects

Ctrl+ scroll tabs values -/+

Shift+space **tools** — ADD AN EMPTY GREASE PENCIL TO ACTIVATE DRAW

reset value — MESSED UP WITH OPTIONS AND VALUES? RESET TO DEFAULT...

copy * to selected — COPY INPUT VALUES BETWEEN MULTIPLE OBJECTS

Camera

SET VIEW ON SELECTED CAMERA

Ctrl+0 active **Camera**

Track to Camera follow — TARGET: TO OBJECT / TO: -Z UP: Y

Shift+ fly/walk mode

space gravity

Ctrl+alt+0 **set camera** on current view

edit

edit

Mesh context menù face, edge, vertex

S measures — ANGLES AND DISTANCE

Knife — &x/y/z axis constrain — &C cut through — &a angle constraint — CAN'T BE USED RELATIVE TO EDGES

shift snap on mid

ctrl+shift+X Carver cutting ops — space select tool

Face or edge — create between selected

J Connect vertices path

Mesh tools — T — EDIT > EDGE TOOLS > SET EDGE LENGHT — Specify edge lenght

Shift+Ctrl+B **ool** tool — BOOLEAN OPS MADE EASY — Ctrl+ union / differ

V **Rip** vertices or edges

Y **Split** selected geometry

Merge — REMOVE DOUBLE VERTICES

Auto merge — TRANFORM ORIGIN

Split edges and faces — AUTO CORRECT LOTS OF MANIFOLDS WHILE MODELING

f3 Mesh bisect — **Bisect** with a plane

Intersect (knife)

Intersect (boolean)

Intersect selected

Extrude — &x/y/z on axis — alt+E EXTRUDE MODES DIALOG — EXTRUDE MANIFOLD? EXTRUDE INWARDS? WITHOUT? LEAVING DOUBLE FACES — click **extrude**

Ctrl+Bevel — faces or edges — &C bevel vertex — segments

SPIN WORKS PERPENDICULAR TO THE VIEW, SET STEPS AND ANGLE — f3+Spin

Inset ×2 individual faces inset — ctrl+ extrude inset

TIP: HIDE OTHER FACES TO LIMIT LOOPCUT — Ctrl+R **Loopcut** and slide — number of cuts — &e even — MATCHES THE SHAPE OF THE ADJACENT EDGE LOOP — &flip RUN THE TARGET EDGE LOOP TO MATCH

Ctrl+R **Offset** edge slide

alt+N **ormals** menu — Shift+N recalc **normals**

Unwrap

Mesh cleanup — MESH > CLEAN UP > LIMITED DISSOLVE — Remesh — VOXEL / QUADS

Optimize mesh — Remove un-needed vertices — Shift+G — Amount of connecting edges — AND DISSOLVE WITH CTRL+X

Ctrl+Join selected

P **Separate** selected from mesh

proportional edit O — Shift+ FALLOFF CURVE

SUZANNE WITH FEDORA

Edge data

UV SEAM · BEVEL WEIGHT · SHARP · CREASE · FREE STYLE

organize

f2 **Rename** object — Ctrl+f2 BATCH RENAME

Move to collection

Shift+ Link to collection

scenes · collections · render layers · object · compositor · data

Create collection — HIDE INDIRECT ONLY · MASK OUT · HIDE PRODUCTION · HIDE RENDER

Ctrl+P **arent** element

Alt+P Clear **arent**

Ctrl+L make links from/active to selected — TRANSFER PROPERTIES (MATERIALS, UV MAPS, GROUPS, ANIM DATA, MODIFIERS...)

Shift+A — INSERT MULTI OBJECT — COLLECTION INSTANCE

Ctrl+A MAKE INSTANCES REAL — TO SET INSTANCED COLLECTION OBJECTS TWO UNDERNEATH > COLLECTION > OBJECT — RECONVERT COLLECTION TO INSTANCE — Lorem ipsum

Ctrl+G **Vertex groups** — CREATE, ASSIGN, REMOVE, SET

Add to Asset browser — = Mark as asset — Link, Append, A. (reuse data) — ADD ID DATA TO YOUR LIBRARY AND LINK OR APPEND LATER TO PROJECTS — ACTIONS, OBJECTS, MATERIALS, NODE TREES, WORLD

modifiers

Subdivision surface Increase mesh detail. Use together with smooth shading and edge crease to obtain perfect detailed objects.

Auto Smooth — ACTIVATE SMOOTH SHADING AND FIX BLACK+EDGES PROBLEMS IN RENDERING

Array Duplicate a series of objects dynamically.

Skin create an armature around the edge. Perfect for branches-like objects.

Ctrl+A Radius of the selected vertices

Shrinkwrap shrink the mesh to the target object — TIP: USE IT TO RETOPOLOGIZE A MESH

Bevel Bevel edges dynamically.

Solidify Adds depth to a flat surface.

Decimate Optimize a mesh and reduce polygons.

Multiresolution Increase mesh detail dynamically for detail sculpting

nodes

NODES WORKS IN SIMILAR WAYS FOR THE SHADER, COMPOSITOR AND GEOMETRY NODES EDITORS

F **Connect** selected nodes

Ctrl+G **roup** selected nodes — Ctrl+alt+G Ungroup — tab Edit group

Shift+ Add reroute

Alt+S witch input

M **ute** node

Ctrl+ **ide** unused sockets — SIMPLIFY NODE MAPS REMEMBER TO UNHIDE IF NEEDED!

shaders

NODE WRANGLER

Ctrl+Shift+ **Connect** to material output — PREVIEW SELECTED NODE TO OUTPUT

Shift+W node wrangler context menu — Ctrl+T add texture setup

Shift+Q **Material** utils — MORE SELECT, REMIX, OVERRIDE

Shift+ delete material

Basic material PBR node example — PBR: PHYSICALLY BASED RENDERING

TEXTURE COORD · MAPPING · IMAGE TEXTURE · MIX RGB · PRINCIPLED BSDF · MATERIAL OUTPUT · NORMAL MAP · BUMP · COLOR RAMP · DISPLACEMENT

color · vector · factor · shader

RGB MIXES | SHADER TWEAKS

Micro displacement

Toon node example — BSDF · SHADER TO RGB · COLOR RAMP · MATERIAL OUTPUT

COLOR PICKER EVERYTHING TO THE INPUT!

geometry nodes

GROUP INPUT — DO SOME THINGS — GROUP OUTPUT

Very simple object transform

GROUP INPUT — TRANSFORM GEOMETRY — GROUP OUTPUT — MOVE, ROTATE OR SCALE

Snap

Shift+tab **Snap**

Ctrl+Shift+tab **Snapping** type

align rotation · absolute grid align

increment · volume · vertex · edge center · edge · edge perp. · face

set snap base — &b — &a — TARGET ANYWHERE LOCATION — add snap point — &alt+ rotate view while snapping

paint

Shift+X **Color** picker — add color swatch

Move stencil — Shift+ scale stencil — Ctrl+ rotate stencil

4 X **Flip** colors

Ctrl+X **Set** vertex color

grease pencil

2D

File > New > 2d anim — DEDICATED WORKSPACE. WHITE BOARD AND 2D FRIENDLY WORKSPACE

HOLD — Free hand draw

Ctrl+ Erase

Shift+alt+ Polygon draw

alt+ stroke select

stroke depth order — DEPTH HIERARCHY

Object > Converto to > Grease pencil — FROM MESH TO GREASE PENCIL

File > export > grease pencil as svg — SAVE VECTOR FROM GREASE PENCIL

sculpt

Brush

F Size

Shift+ strength

Ctrl+ rotation

X flip colors

Add

Ctrl+ Subtract

Shift+ **Soft**

M Mask paint

alt+M ask remove

alt+E **Stroke** method

Remesh

Voxel

Quadriflow

Texture baking

- Create a new UV Unwrap
- Add an empty image texture to each material, without connecting to shader output
- Select target image texture node and UV map node
- Set bake settings (common settings: diffuse, no direct and indirect light)
- Bake and save image
- Repeat for each map type (color, roughness, normal...)
- Check "selected to active" to bake a mesh on top of another

Render f12

Ctrl+f12 Render animation

view > viewport **Render** — RENDER ANY VIEWPORT WITHOUT SETTING THE CAMERA. VERY USEFUL, ESPECIALLY WITH REVIEW!

f11 show last render

Ctrl+Border render

Ctrl+alt+Border render

Shift+Save image

Export transparent video — FILE FORMAT: FFMPEG CONTAINER: QUICKTIME CODEC: QT RLE

Export render layers in a single image

compositor

Still life with shadow on separate layers — BASIC NODE SETTINGS SAVING A SINGLE IMAGE FILE WITH A SEPARATE TRANSPARENT SHADOW LAYER

RENDER LAYERS · ALPHA OVER · VIEWER · COMPOSITE · SET ALPHA · COLOR RAMP · FILE OUTPUT · N

SAVE TO OPEN EXR MULTILAYER. REMEMBER ORDER IS BY FILE OUTPUT INPUT NAME. ACTIVATE DWAA COMPRESSION (LOSSY BUT UNDISTINGUISHIBLE)

addons

Node wrangler: Material node previews and more. **Looptools:** bridge faces, flatten faces, convert faces to circle shape, distribute vertex, etc. **Mesh Tools:** several tools to aid modelling **Booltool:** manage non destructive boolean modifier ops. **Material Utilities:** context menù, assign and manage materials; selections, slots, overrides. **Copy Attributes Menù:** copy from active to selected. **Import Images as Plane:** hang a painting in a click. **Extra Objects** (mesh and curves): add gears, diamonds, pipes, single vertor the old dear teapot (Shift+A). **Add Curve sapling:** Tree generator. **Bsurface:** create surfaces with grease pencil. **Carver:** cuts and holes made easy. **Archimesh:** generate architecture dynamical elements (rooms, doors, windows, columns, stairs, tile roofs). **Measureit ARCH:** measure distances, angles, surfaces and areas. And render with labels, quotes and cad style helpers.

animation

space PLAY

Insert keyframe

alt+ Remove keyframe

prev. frame · next frame

Light and material properties

Reflection · Specular · Roughness/bump · Fresnel · Anisotropy · Dicroic · Muty layer · Diffuse · Ambient · Scattering · Diffraction · Emission · Transparency · Refraction · Caustics · Dispersion · Translucency

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